

Statement by Patent (and Patent Application) Owner(s)

(Please check one of the following two options.)

☐ I, (full name) , (full postal address) , am the owner of the patent/patent application (patent number/patent application number) ;

☒ I, Long Chen , 4# Zhongguancun South Fourth Street, Haidian District, Beijing, China, 100190, am the authorized representative of the owner Institute of Software Chinese Academy of Sciences of the patent/patent application CN201711379239.4, CN202310632536.4, CN202410875797.3, and PCT/CN2024/104916 (International Publication No. WO2026007162A1, the third of which is its Chinese family patent),

and do hereby agree to grant to any interested party, including the program organizer, public reviewers, etc., the right to freely use the above patent (and patent application) for evaluation purpose, and commit to grant to any interested party, including standard drafter, standard organization department, standard users, etc., a public, irrevocable, nonexclusive, royalty-free license for using the patent (and patent application) without any conditions, if the algorithm Loom (Proposal No.: PKCKEX-733110), Weaver (Proposal No.: PKCKEM-568232), and Shuttle (Proposal No.: PKCSIGN-729909) is selected for standardization.

Signed: _____

Date: _____

Long Chen

2026.6.28

POWER OF ATTORNEY

Principal (Sole and Lawful Owner of the Patent/Patent Application):

Name: Institute of Software, Chinese Academy of Sciences

Unified Social Credit Code: 121000004000123696

Legal Representative/Person in Charge: Zhao Chen

Address: No. 4 Zhongguancun South Fourth Street, Haidian District, Beijing

Contact Phone: 010-62661068

Attorney (Authorized Representative):

Name: Chen Long

ID Number: 340111198804054514

Employer and Position: Associate Researcher, Institute of Software, Chinese Academy of Sciences

Contact Phone: 18810460083

Mailing Address: No. 4 South Fourth Street, Haidian District, Beijing

I. Scope of Authorization and Matters Entrusted:

The Principal is the sole, complete, and lawful owner of the patents/patent applications listed in the attachment to this Power of Attorney, with clear and unencumbered title, free of any pledge, exclusive license, or ownership dispute. The Principal hereby specially authorizes the Attorney to act as the Principal's authorized representative, within the scope and term expressly limited by this Power of Attorney, solely to handle intellectual property declaration matters related to the New Generation Commercial Cryptography Algorithm Global Collection Activity (NGCC) organized by the Commercial Cryptography Standardization Research Institute. The specific scope of authorization is strictly limited to the following:

1. To sign on behalf of the Principal the "Patent Owner and Patent Applicant Declaration" in Appendix B.2 of the "Public Key Cryptographic Algorithm Submission

Requirements" officially released for this collection activity; if the Attorney is required to sign any other supporting legal documents directly related to the intellectual property matters of the submitted algorithm for this collection activity, prior written review and confirmation with the Principal's official seal is required, and the Attorney shall not sign any other documents without prior written confirmation; any unauthorized signing beyond the scope of authority without the Principal's written confirmation shall not have any legal binding force on the Principal from the outset;

2. To make, on behalf of the Principal and with respect to the patents/patent applications listed in the attachment to this Power of Attorney, confirmations and commitments fully consistent with the official requirements of this collection activity: namely, if the corresponding algorithm is selected as a winning algorithm in this collection activity and proceeds to national standard formulation, the Principal shall grant to standard drafters, standard administration departments, standard users, and other relevant parties, for the purpose of implementing the standardized algorithm, a worldwide, unconditional, public, irrevocable, non-exclusive, and royalty-free patent/patent application implementation license for the entire life cycle of the national standard, in accordance with the "Public Key Cryptographic Algorithm Submission Requirements". The Attorney has no authority to expand the scope of the license, increase licensing obligations, change licensing restrictions, or make any commitments beyond the foregoing scope;

3. To handle document submission, official content interpretation, and supporting procedural formalities directly related to the authorized matters in Items 1 and 2 above; without separate written special authorization with the Principal's official seal, the Attorney has no authority to make any substantive commitments regarding rights and obligations, change the Principal's intellectual property ownership and interests, or dispose of any of the Principal's intellectual property.

II. List of Patents/Patent Applications Covered:

The patents/patent applications covered by this Power of Attorney are detailed in Annex 1 "Patent/Patent Application Information Detail Table", which constitutes an integral and inseparable part of this Power of Attorney.

III. Term of Authorization:

This Power of Attorney shall be effective from the date of signing until the later of the following dates:

(1) The date on which all processes of this New Generation Commercial Cryptography Algorithm Collection Activity (including multiple rounds of evaluation and standardization of winning algorithms) are fully completed;

(2) The date on which all obligations under the relevant documents signed under the authorization of this Power of Attorney have been fully performed.

Upon expiration of the validity period, this authorization shall automatically become invalid.

IV. Prohibition of Sub-delegation:

The Attorney shall personally handle the entrusted matters under this Power of Attorney. Without the Principal's prior written consent with official seal, the Attorney shall not sub-delegate any rights or obligations under this Power of Attorney to any third party.

V. Liability and Validity:

(1) The Attorney shall strictly exercise authority within the scope and term expressly limited by this Power of Attorney, abide by the principle of good faith, and shall not act beyond the scope of authority. Any act exceeding the scope of authorization under this Power of Attorney does not represent the true intent of the Principal, and all legal liabilities arising therefrom shall be borne solely by the Attorney; the Principal shall not bear any liability.

(2) The Principal shall bear legal liability only for the Attorney's signing acts within the scope and term expressly limited by this Power of Attorney that comply with the official requirements of this collection activity. This delegation is a special and limited authorization and does not constitute a general authorization, full authorization, or continuing general authorization to the Attorney.

(3) Any modification or change to this Power of Attorney shall be effective only if both the Principal and the Attorney sign a written document.

VI. Commitments of Both Parties:

(1) Principal's Commitment:

The Principal recognizes and shall bear corresponding legal liability for relevant documents signed by the Attorney within the scope and term expressly limited by this Power of Attorney that comply with the official requirements of this collection activity.

(2) Attorney's Commitment:

The Attorney shall strictly comply with all provisions of this Power of Attorney and shall not make any commitments or take any actions beyond the scope of authorization. If the Attorney violates the provisions of this Power of Attorney or acts beyond the scope of authority, causing loss to the Principal, the Attorney shall bear full compensation liability and corresponding legal consequences.

VII. Annexes:

Annex 1: "Patent/Patent Application Information Detail Table"

Annex 2: Commercial Cryptography Standardization Research Institute "Public Key Cryptographic Algorithm Submission Requirements" Appendix B.2 "Patent Owner and Patent Applicant Declaration" (Standard Text to be Signed)

Principal (Official Seal):



Legal Representative/Person in Charge (Signature):

Attorney (Signature):

Long Chen

Date: 18 Year 2026 Month 6 Day

Annex 1: Patent/Patent Application Information Detail Table

No.	Patent Name	Patent Application No.	Inventor(s)	Patent Status
1	A Post-Quantum Key Agreement Method	CN201711379239.4	Zhang Zhenfeng; Chen Long; Wang Ke	Granted
2	A Post-Quantum Key Encapsulation Method and System Applicable to TLS Protocol	CN202310632536.4	Chen Long; Zhang Zhenfeng; Zhuang Zidi; Han Jiang	Substantive Examination
3	A Method and System for Constructing AKE Protocol from PKE Algorithm with Errors	CN202410875797.3	Chen Long; Zhang Zhenfeng; Zhuang Zidi	Substantive Examination
4	A Method and System for Constructing AKE Protocol from PKE Algorithm with Errors	PCT/CN2024/104916 (International Publication No. WO2026007162A1; the 3rd family member is the Chinese counterpart patent)	Chen, Long; Zhang, Zhenfeng; Zhuang, Zidi	Not Entered National Phase - Within PCT Validity Period

Annex 2

Commercial Cryptography Standardization Research Institute
"Public Key Cryptographic Algorithm Submission Requirements"
Appendix B.2 "Patent Owner and Patent Applicant Declaration"

(Standard Text to be Signed)

B.3 Statement by Reference and Optimized Implementations' Owner(s)

(Please check one of the following two options.)

- ☐ I, (full name), (full postal address), am the owner of the submitted reference implementation and optimized implementations of the algorithm (name of algorithm) ;
- ☐ I, (full name), (full postal address), am the authorized representative of the owner (full name) of the submitted reference implementation and optimized implementations of the algorithm (name of algorithm) .

and do hereby agree that any interested party, including the program organizer, public reviewers, etc., can freely use such implementations for evaluation purpose, and commit to place no restriction on worldwide disclosure and free adoption if the algorithm is selected for standardization.

Signed:

Date:

(Note: If there are multiple owners of the reference and optimized implementations, the statements shall be signed by each and every one of them or their authorized representatives. The authorized representative shall submit the letter of authorization.)